

RUI DING

AUTONOMOUS MATERIALS DISCOVERY | SCIENTIFIC AGENTS & DEVICE TWINS CLOSED-LOOP EXPERIMENTATION

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PROFESSIONAL PROFILE

End-to-end autonomous discovery scientist combining AI systems engineering with hands-on materials experimentation. Leads DToR, T³, and RAPIDS and develops of the NSF MADE-PUBLIC data platform (FMRG #2037026). Builds scientific-agent pipelines, material-device digital twins, reliability-aware simulation workflows, and closed-loop discovery systems. Experience spans data mining, ML modeling, HPC workflow orchestration, DFT/MD/MLIP verification, nanomaterial synthesis, electrochemical characterization, and device fabrication. Demonstrated closed-loop discovery through a five-round, 258-sample campaign that reduced synthesis failure from 47% to zero.

1,836 citations (Aug 2026) | h-index 24 | 16 first- or co-first-author peer-reviewed papers

CORE TECHNICAL EXPERTISE

Autonomous discovery & scientific agents: multi-agent orchestration; planning and sequential decision-making; RAG; tool calling; state inspection and structured logging; uncertainty- and reliability-aware routing; active learning; Bayesian optimization; provenance; benchmark design; and human-in-the-loop evaluation.

Scientific ML, modeling & experiment: classical machine learning; graph neural networks; spiking neural networks; explainable and multi-fidelity learning; DFT/AIMD (VASP, QE); MD (GROMACS); ML interatomic potentials; ORCA/Gaussian; COMSOL; nanomaterial and electrocatalyst synthesis; electrochemical characterization; device fabrication; and field-effect-transistor (FET) sensing.

Research software, data & HPC: Python and scientific/ML libraries; PyTorch; PyTorch Geometric; Transformers; LangGraph/LangChain; local/open-weight LLM inference; Chroma; SQLite; Pydantic; Docker; Linux; Git; Django/Flask; HTML/CSS/JavaScript; Globus; scientific visualization; and SLURM/PBS and HPC workflows.

PROFESSIONAL APPOINTMENTS

Eric and Wendy Schmidt AI in Science Postdoctoral Fellow, University of Chicago. Advisors: Prof. Sep 2023 - Present
Junhong Chen (PME/Argonne) and Prof. Yuxin Chen (Computer Science).

Resident Associate, Chemical Sciences and Engineering Division, Argonne National Laboratory. Sep 2023 - Present
Host: Prof. Junhong Chen; autonomous materials and device discovery spanning AI/ML agents, nanomaterials, electrochemistry, sensors, atomistic simulation, and experimental validation; user of ALCF and the Center for Nanoscale Materials.

EDUCATION

Ph.D. (Doctor of Engineering), Materials Science and Engineering, Nanjing University, Nanjing, China. Research: machine learning, electrocatalysts, membrane-electrode assemblies, fuel cells, and water electrolyzers. Sep 2018 - Jun 2023

Visiting Graduate Researcher, Department of Chemical and Biological Engineering, Hong Kong University of Science and Technology. Oct 2022 - Apr 2023

B.Eng., Renewable Energy Science and Engineering, Nanjing University, Nanjing, China. Sep 2014 - Jun 2018

SELECTED AUTONOMOUS DISCOVERY AND AI SYSTEMS IMPACT

- **Closed-loop experimental materials discovery.** Led a five-round campaign spanning 534 publications, active learning, domain-adapted DFT, and synthesis/electrochemical testing of 258 multimetallic oxides; synthesized and tested candidates across all five rounds. Drove synthesis failure from 47% to zero during five iterative ML model guided batches guided and the best acidic-OER overpotential at 10 mA cm⁻² from 209 to 154 mV, discovering a Ru-Mn-Ca-Pr oxide that outperformed commercial IrO₂ at 1 A cm⁻² and sustained 125 h in a proton exchange membrane (PEM) electrolyzer ([Science Advances](#), 2025).
- **Deep Tree of Research (DToR)** [[GitHub](#)]. Designed and led a locally deployable LangGraph multi-agent system for evidence-grounded scientific planning, with explicit expand-prune-finalize routing, checkpointed execution, multiple model/search backends, state inspection, structured logs, Docker deployment, and SLURM/PBS orchestration. Benchmarked across 27 materials/device topics against 44 agentic systems using rubric scoring and 14,580 pairwise comparisons; one DToR hypothesis helped guide the wet-lab fabrication of a selective short-chain per- and polyfluoroalkyl substances (PFAS) FET sensor. AI4Mat@NeurIPS 2025.
- **Text-Twin-Translation (T³)** [[GitHub](#)]. Led the full-stack development of T³: TextGrad-optimized extraction of 28-field device knowledge graphs from more than 1,600 FET sensor papers, a topology-aware PyTorch/PyG digital twin achieving up to 92.3% prediction accuracy, and DFT/MD-validated screening of 123.2 million molecular candidates. Accepted to [KDD 2026's AI for Sciences Track](#); Spotlight Oral at AI4Mat@ICLR 2026.

- **RAPIDS** [\[GitHub\]](#). Built an agent-ready, reliability-aware atomistic verification system with configurable pose generation, guarded relaxation, automatic structure validation, pluggable MLIP backbones, browser access, an MCP server, and escalation to ORCA or GPU4PySCF for DFT simulation. RAPIDS-Select routes only unreliable cases to higher fidelity under a cost-aware policy. Across 5,567 probe-target complexes and nine fidelity arms, RAPIDS achieved 1.37 kcal mol⁻¹ error on covered chemistry while cutting cost by ~39% at matched accuracy. AI4Physics@ICML 2026.
- **NSF MADE-PUBLIC data platform (FMRG #2037026)**. Designer/developer of the platform's browser-accessible infrastructure layer, supporting Globus-based data lifecycle management, searchable metadata, interactive knowledge graphs, Bayesian experiment design, data-driven screening, and atomistic-simulation modules.

RESEARCH SUPPORT AND COMPUTING AWARDS

AI in Science Schmidt Fellowship Seed Fund (\$33,000) - Principal Investigator. AI-Guided Discovery of PFAS-Selective Membrane Interfaces: From Autonomous Design to Experimental Validation.	2026
Science Labs that Only AI Can Build - ML Lead. Text-Twin-Translation: AI-guided discovery of robust PFAS chemical FET sensors and bio-FET platforms in real matrices.	2025
NSF Supplemental Award (FMRG 2037026 and DMREF 2323296) - ML Lead. ML-driven integration of 2D materials for printable sensors in biosensing and environmental monitoring.	2025
AI+Science Research Initiative (\$5,000) - Principal Investigator. BRAINIAC: AI-driven optimization of nano-micro materials and devices.	2024
Computing and user-facility awards: NERSC Exploratory Allocation ERCAP0039011 / m5358 (2026); NSF ACCESS MAT240097 (2024-2026); Argonne Leadership Computing Facility ID16430 (2024); Center for Nanoscale Materials Service Unit Award (2024).	

PUBLICATIONS

Selected first- and co-first-author publications and papers; 16 peer-reviewed in total. **Rui Ding** is bolded; * denotes equal contribution. Additional coauthored work is available on Google Scholar.

Archival Conference Paper

- [C1] Rui Ding***, Zixin Ding*, Rodrigo P. Ferreira*, Yuxin Chen, and Junhong Chen. Text-Twin-Translation (T³): A Full-Stack Machine Learning Framework for Functional Material-Device Systems Discovery. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), AI for Sciences Track, main conference, 2026 (Preliminary version selected for a Spotlight Oral at AI4Mat@ICLR 2026) [DOI](#)

Peer-Reviewed Journal Articles

- [J1] Rui Ding**, Jianguo Liu, Kang Hua, Xuebin Wang, Xiaoben Zhang, Minhua Shao, Yuxin Chen, and Junhong Chen. Leveraging Data Mining, Active Learning, and Domain Adaptation in a Multi-Stage, Machine Learning-Driven Approach for the Efficient Discovery of Advanced Acidic Oxygen Evolution Electrocatalysts. *Science Advances* 11, eadr9038 (2025). [DOI](#)
- [J2] Rui Ding**, Junhong Chen, Yuxin Chen, Jianguo Liu, Yoshio Bando, and Xuebin Wang. Unlocking the Potential: Machine Learning Applications in Electrocatalyst Design for Electrochemical Hydrogen Energy Transformation. *Chemical Society Reviews* 53, 11390-11461 (2024) (Journal cover). [DOI](#)
- [J3] Rodrigo P. Ferreira***, **Rui Ding***, Fengxue Zhang, Haihui Pu, Claire Donnat, Yuxin Chen, and Junhong Chen. Expediting Field-Effect Transistor Chemical Sensor Design with Neuromorphic Spiking Graph Neural Networks. *Molecular Systems Design & Engineering* 10, 345-356 (2025) (Journal cover). [DOI](#)
- [J4] Rui Ding**, Ran Wang, Yiqin Ding, Wenjuan Yin, Yide Liu, Jia Li, and Jianguo Liu. Designing AI-Aided Analysis and Prediction Models for Nonprecious Metal Electrocatalyst-Based Proton Exchange Membrane Fuel Cells. *Angewandte Chemie International Edition* 59, 19175-19183 (2020). [DOI](#)
- [J5] Rui Ding**, Xuebin Wang, Aidong Tan, Jia Li, and Jianguo Liu. Unlocking New Insights for Electrocatalyst Design: A Unique Data Science Workflow Leveraging Internet-Sourced Big Data. *ACS Catalysis* 13, 13267-13281 (2023). [DOI](#)
- [J6] Rui Ding**, Yawen Chen, Ping Chen, Ran Wang, Jiankang Wang, Yiqin Ding, Wenjuan Yin, Yide Liu, Jia Li, and Jianguo Liu. Machine Learning-Guided Discovery of Underlying Decisive Factors and New Mechanisms for the Design of Nonprecious Metal Electrocatalysts. *ACS Catalysis* 11, 9798-9808 (2021). [DOI](#)
- [J7] Rui Ding**, Yiqin Ding, Hongyu Zhang, Ran Wang, Zihan Xu, Yide Liu, Wenjuan Yin, Jiankang Wang, Jia Li, and Jianguo Liu. Applying Machine Learning to Boost the Development of High-Performance Membrane Electrode Assembly for Proton Exchange Membrane Fuel Cells. *Journal of Materials Chemistry A* 9, 6841-6850 (2021) (Inside cover). [DOI](#)
- [J8] Rui Ding**, Meng Ma, Yawen Chen, Xuebin Wang, Jia Li, Guoxiong Wang, and Jianguo Liu. Inspecting Design Rules of Metal-Nitrogen-Carbon Catalysts for Electrochemical CO₂ Reduction Reaction: From a Data Science Perspective. *Nano Research* 16, 264-280 (2023). [DOI](#)
- [J9] Rui Ding**, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Jia Li, Xuebin Wang, and Jianguo Liu. Machine Learning Utilized for the Development of Proton Exchange Membrane Electrolyzers. *Journal of Power Sources* 556, 232389 (2023). [DOI](#)

- [J10] **Rui Ding**, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Guiding the Optimization of Membrane Electrode Assembly in a Proton Exchange Membrane Water Electrolyzer by Machine Learning Modeling and Black-Box Interpretation. *ACS Sustainable Chemistry & Engineering* 10, 4561-4578 (2022). [DOI](#)
- [J11] **Rui Ding**, Shiqiao Zhang, Yawen Chen, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiaoke Li, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Application of Machine Learning in Optimizing Proton Exchange Membrane Fuel Cells: A Review. *Energy and AI* 9, 100170 (2022). [DOI](#)
- [J12] **Rui Ding***, Wenjuan Yin*, Gang Cheng, Yawen Chen, Jiankang Wang, Xuebin Wang, Min Han, Tianren Zhang, Yinliang Cao, Haimin Zhao, Shengping Wang, Jia Li, and Jianguo Liu. Effectively Increasing Pt Utilization Efficiency of the Membrane Electrode Assembly in Proton Exchange Membrane Fuel Cells through Multiparameter Optimization Guided by Machine Learning. *ACS Applied Materials & Interfaces* 14, 8010-8024 (2022). [DOI](#)
- [J13] **Rui Ding**, Yawen Chen, Xiaoke Li, Zhiyan Rui, Kang Hua, Yongkang Wu, Xiao Duan, Xuebin Wang, Jia Li, and Jianguo Liu. Atomically Dispersed, Low-Coordinate Co-N Sites on Carbon Nanotubes as Inexpensive and Efficient Electrocatalysts for Hydrogen Evolution. *Small* 18, 2105335 (2022). [DOI](#)
- [J14] **Rui Ding***, Wenjuan Yin*, Gang Cheng, Yawen Chen, Jiankang Wang, Ran Wang, Zhiyan Rui, Jia Li, and Jianguo Liu. Boosting the Optimization of Membrane Electrode Assembly in Proton Exchange Membrane Fuel Cells Guided by Explainable Artificial Intelligence. *Energy and AI* 5, 100098 (2021). [DOI](#)
- [J15] **Rui Ding**, Yide Liu, Zhiyan Rui, Jia Li, Jianguo Liu, and Zhigang Zou. Facile Grafting Strategy Synthesis of Single-Atom Electrocatalyst with Enhanced ORR Performance. *Nano Research* 13, 1519-1526 (2020) (Back cover). [DOI](#)

Preprints and Peer-Reviewed Workshop Papers

- [W1] **Rui Ding***, Zixin Ding*, Rodrigo P. Ferreira*, Yuxin Chen, and Junhong Chen. Geometry, Not Energy Surface, Drives the Neutral MLIP-DFT Gap in Atomistic Interaction Surrogates. *AI4Physics*, at ICML 2026
- [W2] **Rui Ding**, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Hierarchical Deep Research with Local-Web RAG: Toward Automated System-Level Materials Discovery. arXiv preprint (2025); preliminary version at *AI4Mat*, at NeurIPS 2025. [arXiv:2511.18303](#)
- [W3] Rodrigo P. Ferreira*, **Rui Ding***, Rapti Ghosh, Haihui Pu, and Junhong Chen. Neuromorphic Random Walk for Experimental Phosphate Adsorption Modeling. *Machine Learning and the Physical Sciences (ML4PS)*, at NeurIPS 2025

Manuscripts Under Review

- [U1] **Rui Ding**, Haihui Pu, Kuldip Kumar, Jonathan S. Grabowy, Dominic A. Brose, Maria Chan, Seth B. Darling, Andrew Ferguson, Yuxin Chen, and Junhong Chen. From Laboratory Promise to Field Readiness: AI/ML for Discovering Water Materials That Survive Real Water. *Under review at Nature Water*
- [U2] **Rui Ding**, Peng Gong, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Hierarchical Deep Research with Local-Web RAG: Toward Automated System-Level Materials Discovery. *Under review at Nature Machine Intelligence* (Extended manuscript based on W2)
- [U3] **Rui Ding**, Zixin Ding, Rodrigo P. Ferreira, Yuxin Chen, and Junhong Chen. Reliability-Aware Routing for Foundation Atomistic Simulators in Non-Covalent Binding. *Under review at AAAI 2027* (Extended manuscript based on W1)
- [U4] Haihui Pu*, **Rui Ding***, Tetyana Ignatova, Eric A. Josephs, Yuxin Chen, Wei Chen, and Junhong Chen. Nano-AI Convergence: Accelerating Materials Discovery and Overcoming Device Variability in Nanotechnology. *Under review at Digital Discovery*.

Manuscripts in Preparation

- [M1] **Rui Ding** et al. Toward Generalizable Device Twins from Million-Scale Scientific Literature Knowledge Graphs. Manuscript in preparation.
- [M2] **Rui Ding** et al. SciKGFM: Domain-Pretrained Foundation Models for Scientific Knowledge Graphs. Manuscript in preparation.

RESEARCH SOFTWARE, DATA INFRASTRUCTURE, AND OPEN SCIENCE

- **DTor engineering and deployment** [\[GitHub\]](#). Engineered stateful LangGraph workflows, Pydantic configuration, local/web retrieval plug-ins, Chroma vector stores, SQLite checkpoints, structured logging, Docker/Compose packaging, Phoenix/OpenInference instrumentation, and SLURM/PBS cluster scripts with wall-time and signal handling.
- **RAPIDS scientific tool service** [\[GitHub\]](#). Built and deployed modular simulation/configuration pipelines, a browser GUI, interactive 3D outputs, JSON/VASP/report export, pluggable model backbones, structure validation, batch screening, DFT escalation, and an MCP server for agent integration.
- **Evaluation harness design**. As technical lead for the 2026 UChicago AI+Science Hackathon multi-agent/knowledge-graph track, defined the task, curated data, built the evaluation system, and judged six teams. Developed a free composite metric that correlated with a four-model LLM-judge rubric applied to GraphRAG-generated answers at $r = 0.94$.
- **Open-source research code**. Built and maintains GitHub public repositories spanning scientific agents, electrocatalyst data mining, device optimization, FET sensing, atomistic verification, and reproducible AI-for-science training materials.

LEADERSHIP, MENTORING, AND COLLABORATION

Faculty Research Mentor - Schmidt AI in Science Faculty Fellowship; mentored Prof. Omolola Ogbolumani (University of Lagos) on human-in-the-loop AI for food-energy-water nexus optimization.	Jan - Jun 2026
Curriculum Developer - NSF Research Traineeship, University of Chicago; created and open-sourced four end-to-end AI/ML modules using published scientific datasets [GitHub] .	2026
Technical Lead - UChicago AI+Science Hackathon multi-agent framework track; supplied source code and datasets, built the evaluation harness, and judged six teams.	2026
Research Mentor - RENEU program; mentored Covenant Amoo on LLM-based knowledge extraction from FET sensor literature.	2025
Undergraduate Team Mentor - UChicago AI+Science Hackathon; guided a four-student team on AI-enabled materials characterization.	2025
Assistant Research Mentor - RENEU program; assisted in mentoring Abdulafeez Olaitan on nanodroplet dynamics under electron-beam irradiation.	2024
Host Representative - AI+Science Schmidt Fellows Speaker Series, University of Chicago.	2023 - 2026

SELECTED PRESENTATIONS

2026 - Text-Twin-Translation (T^3), ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Jeju Island, Republic of Korea.

2026 - Text-Twin-Translation (T^3) [Spotlight Oral], AI4Mat at ICLR 2026, Brazil.

2026 - Geometry, Not Energy Surface, Drives the Neutral MLIP-DFT Gap [Poster], AI4Physics at ICML 2026, Seoul, Republic of Korea.

2025 - Hierarchical Deep Research with Local-Web RAG [Poster], AI4Mat at NeurIPS 2025, San Diego, CA.

2025 - Neuromorphic Random Walk for Experimental Phosphate Adsorption Modeling [Poster], ML4PS at NeurIPS 2025, San Diego, CA.

2025 - Application of Versatile Machine Learning Approaches to Facilitate the Discovery of Nanomaterials and Devices in Renewable Energy [Contributed Talk], Gordon Research Seminar on Nanomaterials for Applications in Energy Technology, Ventura, CA.

2025 - Journey to a Nanomaterials/Devices Copilot [Invited Presentation], Knowledge-Guided Machine Learning Workshop, University of Michigan, Ann Arbor, MI.

2024-2025 - Data Platform of the NSF MADE-PUBLIC Project (FMRG #2037026) [Project Presentation], NSF MADE-PUBLIC Annual Retreat, Northwestern University and Iowa State University.

SELECTED DISTINCTIONS AND PROFESSIONAL SERVICE

Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship, Schmidt Sciences, Cohort 2. **Sep 2023 - Present**

Spotlight Oral, AI4Mat at ICLR 2026 - Text-Twin-Translation (T^3). **2026**

Journal cover features: Chemical Society Reviews (2024); Molecular Systems Design & Engineering (2025); Journal of Materials Chemistry A (2021); Nano Research (2020).

Conference/workshop reviewer: ACM SIGKDD Main Conference; NeurIPS AI4Mat; ICLR AI4Mat; ICML AI4Physics.

Journal reviewer: ACS Catalysis; Energy & Fuels; Journal of Water Process Engineering; Synthetic Metals; Organic Letters; ACS Applied Energy Materials; Nanomaterials.